

Instrumental variables and 2SLS

Slide route: Lecture9v2.tex: IV, 2SLS, buttery example, cigarette application, exogeneity and relevance. Chapter 12.

Session 12 | Where does identifying variation come from?

Outcomes: State relevance and exclusion/exogeneity, calculate a just-identified IV estimate, explain 2SLS, and recognize weak-instrument concerns.

100-minute route

0–10	Retrieve simultaneity and OVB.
10–30	Instrument logic and the ratio estimator.
30–45	2SLS and the lecture's buttery example.
45–55	Pause and student exclusion critiques.
55–75	General IV model and cigarette application.
75–90	Relevance, weak instruments, exogeneity checks.
90–100	Exit ticket and panel-data transition.

Slide-by-slide script

Slide 1 | Title

Guide Open with the question: where can credible identifying variation come from when treatment is endogenous?

Slide 2 | Roadmap

Guide Preview IV logic, 2SLS, the buttery example, the cigarette application, relevance, and exogeneity.

Slide 3 | Last lesson

Guide Retrieve simultaneity and ask why a strong association can still be a biased causal estimate.

Slide 4 | Instrumental variables

Guide Define an instrument as a source of treatment variation that is relevant and excluded from the outcome equation.

Slide 5 | Instrumental variables

Guide Use the randomized encouragement table: reduced form = $66 - 60 = 6$, first stage = $.50 - .20 = .30$.

Slide 6 | Instrumental variables

Guide Calculate the Wald ratio $6/.30 = 20$ and explain its units.

Slide 7 | Two stages least squares

Guide Describe first-stage prediction and second-stage outcome regression without pretending the manually generated second-stage SE is valid.

Slide 8 | 2SLS

Guide Ask what variation remains in the fitted treatment: variation induced by the instrument and controls.

Slide 9 | Buttery example

Guide Identify the instrument, treatment, outcome, and proposed exclusion restriction before reading the numbers.

Slide 10 | Buttery example

Guide Work the reduced-form and first-stage contrasts on the board.

Slide 11 | Buttery example

Guide Ask students whether the instrument can affect the outcome through taste, income, or another direct channel.

Slide 12 | Buttery example

Guide Connect the ratio calculation to a treatment effect for the variation moved by the instrument.

Slide 13 | Buttery example

Guide State that heterogeneous effects require care: IV need not equal everyone's effect.

Slide 14 | 2SLS model

Guide Write $Y = \beta_0 + \beta_1 D + X'\gamma + u$ and identify the endogenous regressor.

Slide 15 | 2SLS model

Guide Write the first stage $D = \pi_0 + \pi_1 Z + X'\pi + v$ and explain relevance as $\pi_1 \neq 0$ in context.

Slide 16 | Application setup

Guide Before using the cigarette example, name the economic question and the source of exogenous price variation.

Slide 17 | Application

Guide Interpret the IV coefficient in the units of quantity and price, not as a generic correlation.

Slide 18 | Application

Guide Ask whether the proposed instrument shifts demand directly. This is an exclusion question.

Slide 19 | Application

Guide Compare OLS and IV as answers to different identifying assumptions, not as a contest where one must always be larger.

Slide 20 | General IV model

Guide List the maintained assumptions: random sampling, relevance, exclusion/exogeneity, and sufficient variation.

Slide 21 | IV exogeneity

Guide Draw the forbidden direct path from Z to Y and ask students to explain why it invalidates IV.

Slide 22 | IV exogeneity

Guide Emphasize that a strong first stage cannot prove exclusion.

Slide 23 | IV relevance

Guide Explain weak instruments as weak predictive power for the endogenous regressor after controls.

Slide 24 | IV relevance

Guide Ask why weak instruments make estimates noisy and can distort usual large-sample inference.

Slide 25 | IV relevance

Guide Interpret first-stage diagnostics as context-dependent evidence, not a magic cutoff.

Slide 26 | Summary

Guide Have students defend or reject an instrument using separate relevance and exclusion arguments.

Slide 27 | Material

Guide Assign one ratio calculation and one exclusion critique before moving to pooled and panel data.

Worked example | A randomized encouragement

Use with slides Lecture9v2 slides 4–7: Instrumental Variables and Two Stages Least Squares. Use the encouragement table after slide 5 to make the Wald ratio concrete, then connect it to the two-stage explanation on slides 6–7.

Synthetic group means by encouragement Z :

	$Z = 0$	$Z = 1$
Treatment take-up D	.20	.50
Outcome Y	60	66

$$\hat{\beta}_{IV} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}} = \frac{6}{.30} = 20.$$

Explain the units The reduced form is six outcome units per encouragement contrast; the first stage is .30 treatment units per encouragement contrast. The ratio gives 20 outcome units per treatment unit.

Assumptions Encouragement must shift take-up and must not affect outcomes through another channel; random encouragement supports independence. With heterogeneous treatment effects, a local average effect for compliers additionally needs monotonicity and the other usual treatment-design assumptions. Do not call it everyone's treatment effect automatically.

Two stages and the identifying argument

Use with slides Lecture9v2 slides 14–20 for the general IV model, single-endogenous-regressor setup, application framing, and IV assumptions. Return to slides 21–26 when discussing exogeneity, relevance, and weak instruments.

For one endogenous X and exogenous controls W :

$$X_i = \pi_0 + \pi_1 Z_i + \pi'_W W_i + v_i, \quad Y_i = \beta_0 + \beta_1 \hat{X}_i + \beta'_W W_i + \text{residual}.$$

All included exogenous controls belong in the instrument set/first stage. Multiple instruments can predict the endogenous regressor jointly.

Software caution The two stages explain the point estimate. The ordinary second-stage OLS standard errors from manually generated fitted regressors are not valid IV standard errors. Use an IV estimator's covariance calculation.

Illustrative command With appropriately prepared variables:

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ivregress 2sls outcome controls (treatment = instrument), vce(robust)
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The names are placeholders; this is not a command already tested on a supplied dataset.

Weak instruments and exclusion

If the first-stage difference were .02 with the same reduced form of 6, the ratio would be 300. Small denominator variation makes the estimate unstable and conventional approximations unreliable.

First-stage diagnostics Examine excluded-instrument strength conditional on controls. A rule of thumb is not a universal guarantee; design, estimator, and inference assumptions matter.

Exogeneity A strong first stage does not prove exclusion. A transport subsidy that changes both attendance and household disposable income could affect outcomes through multiple channels. Ask students to draw the extra path.

Overidentification With extra instruments, tests can detect certain violations under maintained assumptions. Failure to reject does not certify that all instruments are valid. An exactly identified model has no overidentifying restrictions to test.

Exit answers Relevance concerns association with the endogenous regressor conditional on controls. Exclusion/exogeneity concern the outcome error and direct channels. Strong does not mean valid; robust does not mean weak-IV-proof.

If short / preparation Work the ratio and one empirical story carefully; assign repeated application screenshots. Read the pooled/panel parts of Chapter 10 and the syllabus's Chapter 13 material as relevant.

Source: Lecture9v2.tex; encouragement numbers are synthetic.